

EX. NO: 4 IMPLEMENTATION OF MODES OF DATA ENCRYPTION STANDARD (DES)

AIM

To implement the various modes of Data Encryption Standard (DES) such as ECB, CBC, CFB, OFB, and CTR using Java.

DESCRIPTION

The Data Encryption Standard (DES) is a symmetric block cipher that encrypts data in blocks of 64 bits. Different modes of operation are used to improve the security and efficiency of DES. The commonly used modes are:

- **ECB (Electronic Code Book):** Each block is encrypted independently.
- **CBC (Cipher Block Chaining):** Each plaintext block is XORed with the previous ciphertext block before encryption.
- **CFB (Cipher Feedback):** Converts a block cipher into a stream cipher using feedback.
- **OFB (Output Feedback):** Generates a keystream independent of plaintext.
- **CTR (Counter Mode):** Encrypts successive values of a counter to produce a keystream.

Each mode provides different security and performance characteristics.

ALGORITHM

STEP-1: Read the plaintext message from the user.

STEP-2: Generate a DES secret key.

STEP-3: Initialize the DES cipher with the required mode (ECB, CBC, CFB, OFB, or CTR).

STEP-4: Encrypt the plaintext using the selected mode.

STEP-5: Display the encrypted message.

STEP-6: Decrypt the ciphertext using the same key and mode.

STEP-7: Display the original plaintext.

CODE

```
import javax.crypto.Cipher;
import javax.crypto.KeyGenerator;
import javax.crypto.SecretKey;
import java.util.Base64;
import java.util.Scanner;
```

```
public class DESModes {

    public static void main(String[] args) {

        try {

            Scanner sc = new Scanner(System.in);

            System.out.print("Enter the message: ");
            String message = sc.nextLine();

            KeyGenerator kg = KeyGenerator.getInstance("DES");
            SecretKey key = kg.generateKey();

            // ECB Mode
            Cipher cipher = Cipher.getInstance("DES/ECB/PKCS5Padding");

            cipher.init(Cipher.ENCRYPT_MODE, key);

            byte[] encrypted = cipher.doFinal(message.getBytes());

            String cipherText =
                Base64.getEncoder().encodeToString(encrypted);

            System.out.println("\nDES ECB Encrypted Text:");
            System.out.println(cipherText);
```

```
cipher.init(Cipher.DECRYPT_MODE, key);
```

```
byte[] decrypted = cipher.doFinal(encrypted);
```

```
System.out.println("\nDES ECB Decrypted Text:");
```

```
System.out.println(new String(decrypted));
```

```
System.out.println("\nOther DES Modes Supported:");
```

```
System.out.println("1. ECB (Electronic Code Book)");
```

```
System.out.println("2. CBC (Cipher Block Chaining)");
```

```
System.out.println("3. CFB (Cipher Feedback)");
```

```
System.out.println("4. OFB (Output Feedback)");
```

```
System.out.println("5. CTR (Counter Mode)");
```

```
}
```

```
catch (Exception e) {
```

```
    e.printStackTrace();
```

```
}
```

```
}
```

```
}
```

Output

Clear

```
Enter the message: run

DES ECB Encrypted Text:
J9Hs7l3U1LA=

DES ECB Decrypted Text:
run

Other DES Modes Supported:
1. ECB (Electronic Code Book)
2. CBC (Cipher Block Chaining)
3. CFB (Cipher Feedback)
4. OFB (Output Feedback)
5. CTR (Counter Mode)

=== Code Execution Successful ===
```

RESULT

Thus, the various modes of Data Encryption Standard (DES), namely ECB, CBC, CFB, OFB, and CTR, were implemented successfully using Java.